

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-11 13:30 UTC

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Generated 2026-05-11 13:30 UTC

- Cycles recorded: **603**
- Time span: 423.8h (~1.42 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	566
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	33
Schur-Weyl Duality	17
Representation Theory of Symmetric Groups	17
Free Probability	14
Matroid Theory	12
Additive Combinatorics	10
Polymatroid Theory	8
Algebraic Geometry	7
Random Matrix Theory	7
Noncommutative L^p Geometry	7
Finite Geometry	6
Spectral Graph Theory	6
Plethysm Theory	6
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4

Field	Cycles
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
Free Probability Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3
Diophantine Approximation	3
Persistent Homology	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-11 05:56 UTC	INCONCLUSIVE	Finite-Field Rank Threshold for ACC ⁰ Circuit Complexity
2026-05-11 06:02 UTC	INCONCLUSIVE	Secant Rank Lower Bound for Disjointness Tensors
2026-05-11 06:31 UTC	INCONCLUSIVE	Real Dimension of Parity-Constraint Variety
2026-05-11 07:16 UTC	INCONCLUSIVE	Bounds AC ⁰ Depth
2026-05-11 07:42 UTC	INCONCLUSIVE	Cheeger Constant Inverse Proportional to Extended Frege Proof Siz
2026-05-11 08:00 UTC	INCONCLUSIVE	Communication Matrix Rank Lower Bounds ABP Size for Boolean Funct
2026-05-11 08:32 UTC	INCONCLUSIVE	Invariant Generator Count Bounds SOS Refutation Degree for Symmet
2026-05-11 08:45 UTC	INCONCLUSIVE	Newton Polytope Volume Inverse Proportional to SOS Degree for Max
2026-05-11 10:35 UTC	INCONCLUSIVE	Krull Dimension Lower Bounds SOS Refutation Degree for 3-SAT
2026-05-11 11:01 UTC	INCONCLUSIVE	Finite-Field Rank Threshold for ACC ⁰ Circuit Complexity
2026-05-11 11:36 UTC	INCONCLUSIVE	Gonality Lower Bounds for Disjointness Communication Matrices
2026-05-11 12:25 UTC	INCONCLUSIVE	Projective Plane Disjointness Complexity Bounded by Line Count
		Noncommutative Rank Bounded by Disjointness Communication Complex

Time	Verdict	Title
2026-05-11 12:48 UTC	INCONCLUSIVE	Kronecker Coefficient Asymmetry in Permanent vs Determinant Decom
2026-05-11 13:20 UTC	INCONCLUSIVE	Plethysm Coefficient Growth and Permanent Circuit Complexity
2026-05-11 13:30 UTC	INCONCLUSIVE	Real Radical Rank Lower Bounds SOS Degree for Max-CUT

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```